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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Use the Direct Comparison Test to determine whether the following series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{1}{10n^2 + 5}$$

To determine the convergence or divergence of the series, use the Direct Comparison Test. Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n . If $\sum b_n$ converges, then $\sum a_n$ also converges. If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

Compare the following series.

$$\sum_{n=1}^{\infty} \frac{1}{10n^2 + 5} = \frac{1}{15} + \frac{1}{45} + \frac{1}{95} + \frac{1}{165} + \frac{1}{255} + \frac{1}{365} + \dots$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{36} + \dots$$

Since $10n^2 + 5$ is greater than n^2 for all n , $\frac{1}{10n^2 + 5}$ is less than $\frac{1}{n^2}$ for all n .

Since $n^2 > 0$ for all n , the denominator $10n^2 + 5 > 0$ for all n . The numerator, 1, is always greater than 0. Therefore, $\frac{1}{10n^2 + 5} > 0$. Because $\frac{1}{n^2}$ is greater than $\frac{1}{10n^2 + 5}$, $\frac{1}{n^2} > 0$.

Note that the p-series, $\sum_{n=1}^{\infty} \frac{1}{n^p}$, where p is a real constant, converges if $p > 1$, and diverges if $p \leq 1$. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges because it is a p-series with $p = 2 > 1$.

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is known to be convergent and both series have nonnegative terms. Therefore, by the Direct Comparison Test, the series

$$\sum_{n=1}^{\infty} \frac{1}{10n^2 + 5} \text{ is also convergent.}$$